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 Reuben Kearney

Analytics and Insight

Parameter	Units	Year 0	Year 1	Year 2	Year 3	Year 4
Cotton Gin Calendar Time (Hrs)		5,760	5,760	5,760	5,760	5,760
Cotton Gin Down Time Maintenance (Hrs)		1,434	1,434	1,434	1,434	1,434
Cotton Gin Scheduled Loss (Hrs)		476	476	476	476	476
Cotton Gin Unscheduled Loss (Hrs)		868	868	868	868	868
Cotton Gin Operating Time (Hrs)		130	130	130	130	130
Cotton Gin Available Time (Hrs)		7,256	7,256	7,256	7,256	7,256
Cotton Gin No Scheduled Production (Hrs)		151	151	151	151	151
Cotton Gin Operating Standby (Hrs)		2,389	2,389	2,389	2,389	2,389
Cotton Gin Utilised Time (Hrs)		4,746	4,746	4,746	4,746	4,746
Cotton Gin Operating Delay (Hrs)		1,093	1,093	1,093	1,093	1,093
Cotton Gin Operating Time (Hrs)	hrs	3,653	3,653	3,653	3,653	3,653
Average Metres per Hour	m/hr	54	54	54	54	54
Total Metres Rolled	m	197,253	197,253	197,253	197,253	197,253

Value Driver Modelling – Part 1: What are value driver trees?

📅 June 22, 2015 January 28, 2021 👤 Reuben Kearney

At the peak of the mining boom in Australia it was vogue to use value driver trees to analyse your mining operations and to answer questions like “should I buy more trucks or am I better off with more excavators?”.

These days there’s less consulting dollars floating around for value driver modelling but that hasn’t stopped it from being a fantastic way to visualise and analyse the flow of value from one part of your organisation to another, regardless of your industry.

This is the first post in a series introducing the concept of value drive modelling and providing some practical examples in how to use it.

What ways can VDM be used?

Throughout this series I’ll show you examples of how VDM can;

- identify where the biggest constraints are in an organisation’s ability to create value.
- be used in conjunction with sensitivity analysis to show which areas are at greatest risk for failing to deliver value.
- value a range of different investment options to find the optimal combination for creating value.
- provide transparency at the individual employee level to see how they contribute to the creation of value.
- allow you to benchmark operations that were previously too different to compare through traditional benchmarking methodologies.

What is value?

Before we get to the diagrams, let's start with what we mean by 'value'. Value can be understood as "something of importance, worth, or the usefulness of something". Value can also be described as profit or stakeholder wealth. So, if organisations create value, when and where does this value come from? To visualise this value creation process we can use Porter's value chain (https://web.archive.org/web/20240424065827/https://en.wikipedia.org/wiki/Value_chain).

Figure 1: Porter's Generic Value Chain



A value chain diagram shows a chain of activities for an organisation operating in a specific industry. The chain of activities gives products or services more value than the sum of the independent activities' values. The important distinction here is between primary and support activities. The primary activities are where all the value is directly created. The support activities, while critical for sustaining the business, do not add value directly to the product that the customer ultimately buys.

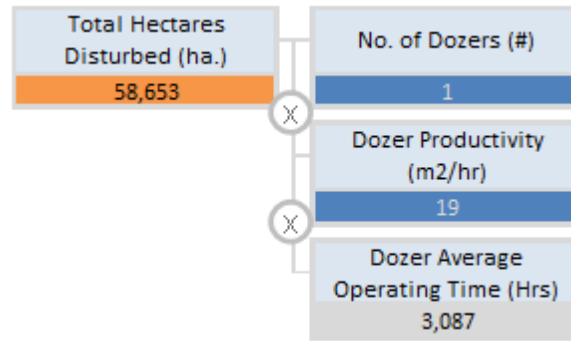
While there are some legitimate criticisms (<https://web.archive.org/web/20240424065827/https://hbr.org/2012/02/why-porters-model-no-longer-wo>) of Porter's value chain (least of all the failure of his consultancy (<https://web.archive.org/web/20240424065827/http://www.forbes.com/sites/stevedenning/2012/11/20/what-killed-michael-porters-monitor-group-the-one-force-that-really-matters/>)) it still provides a straight forward framework for understanding how organisation's might create value.

What are value driver trees?

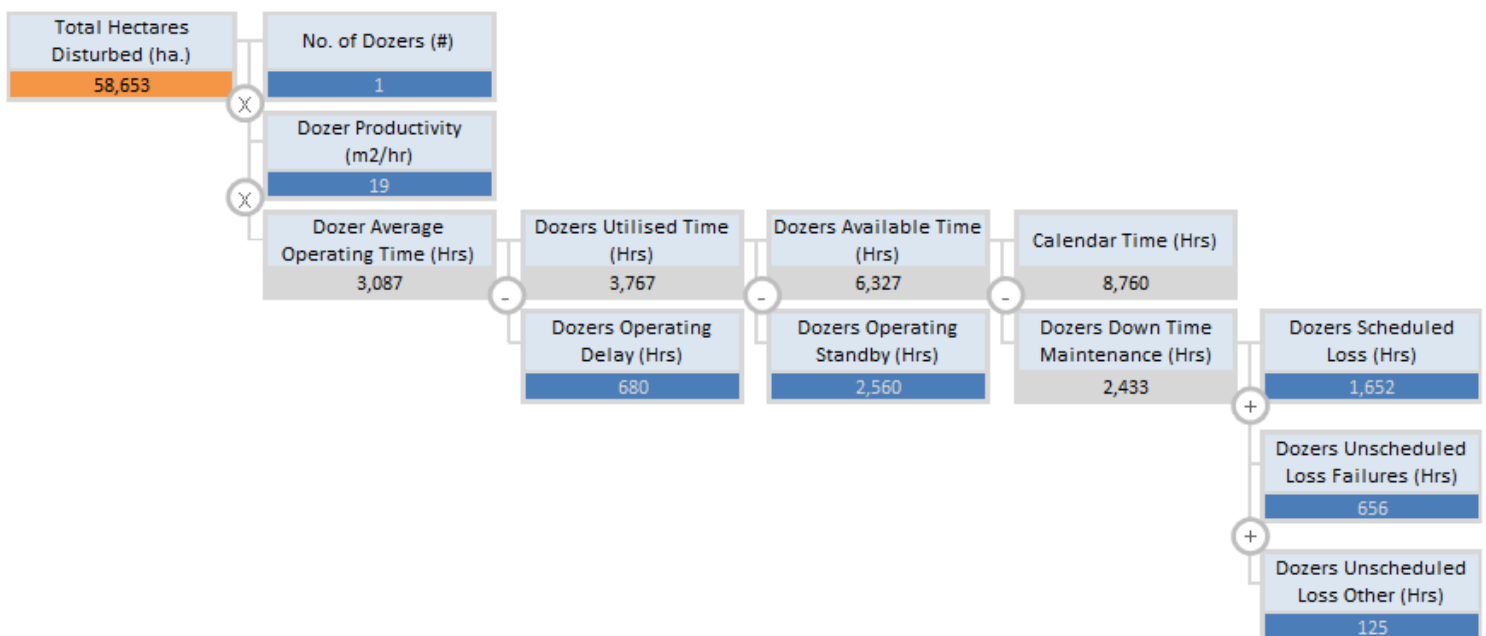
Value Drive Trees (VDTs) are the main type of diagram used as part of VDM. VDTs are basically a picture of the 'gears' or 'value drivers' that power a business. Here we have an example of the basic building block of a VDT.

Total Hectares Disturbed (ha.)
58,653

This building block forms part of a much larger VDT, which in this case is for the productivity of a dozer. The individual elements of this building block are simple. It has a heading, units, and a value. The definition of value in this context, is the amount of hectares cleared per annum.



Now if I expand the tree, we see more elements to the VDT. We see that the boxes are connected in a relationship, and that that relationship is described mathematically, in this case, with multiplication signs. It's apparent that these lower levels elements multiply together to equate to our starting element. This is the fundamental function of a VDT, to transparently show the relationship between different elements of value.



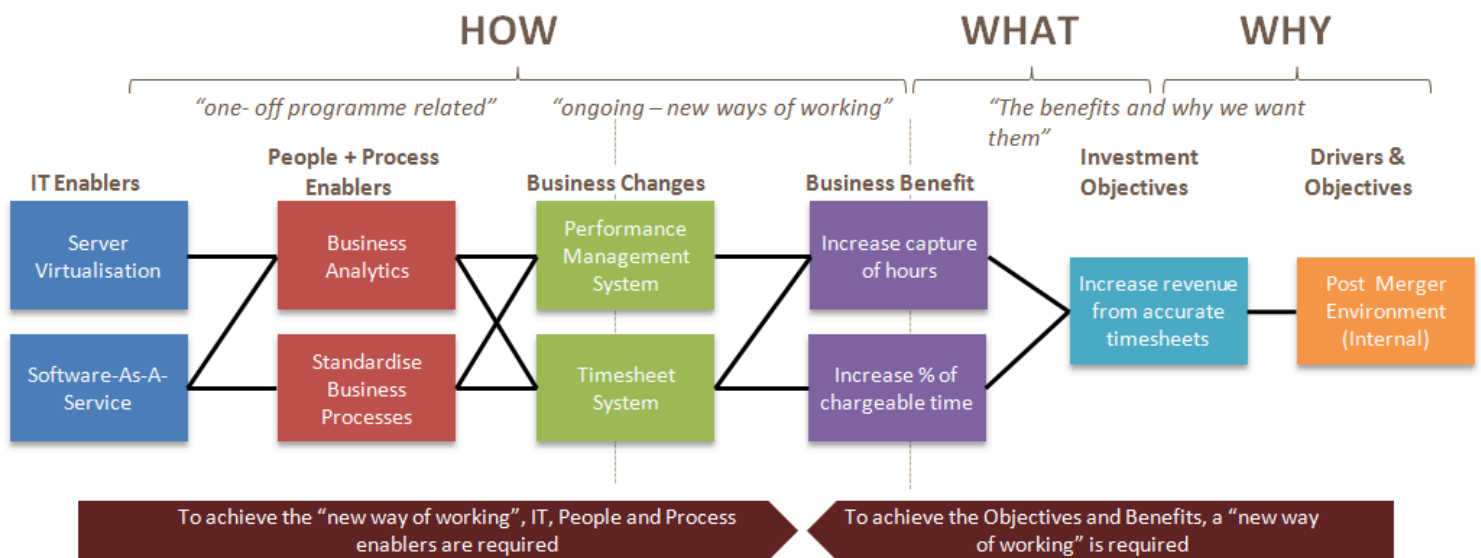
It's possible for this tree to keep breaking down into ever more detailed steps. What's important to remember is that we are interested in those elements that are directly contributing value to the final outcome of the VDT.

What are the different types of VDTs?

The rest of this post will go through the different ways VDTs can be used and visualised. To interact with the VDTs, please download the Excel workbook [here](https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-example-value-driver-trees.xlsx)

(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-example-value-driver-trees.xlsx>).

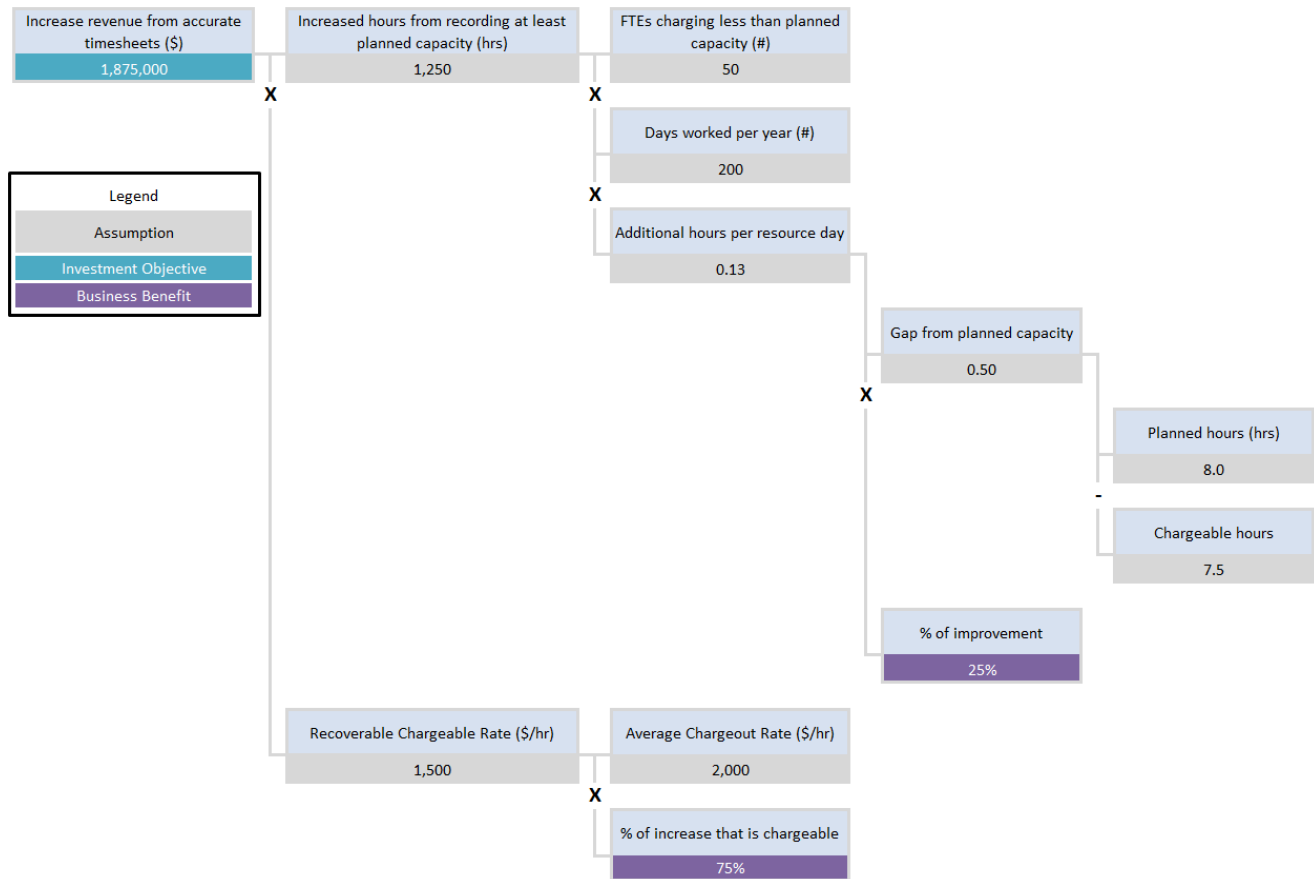
Benefit Realisation VDT



Benefits Realisation VDTs can be built from [Benefits Dependency Networks](https://web.archive.org/web/20240424065827/https://en.wikipedia.org/wiki/Benefits_realisation_management)

(https://web.archive.org/web/20240424065827/https://en.wikipedia.org/wiki/Benefits_realisation_management). A Benefits Dependency Network diagram (as above) shows the inter-relationships between enablers, business changes, benefits and investment objectives.

At its most fundamental level, the purpose of a benefit dependency network diagram is to ensure that you don't double count benefits from interrelated improvements or investments you make to an organisation.

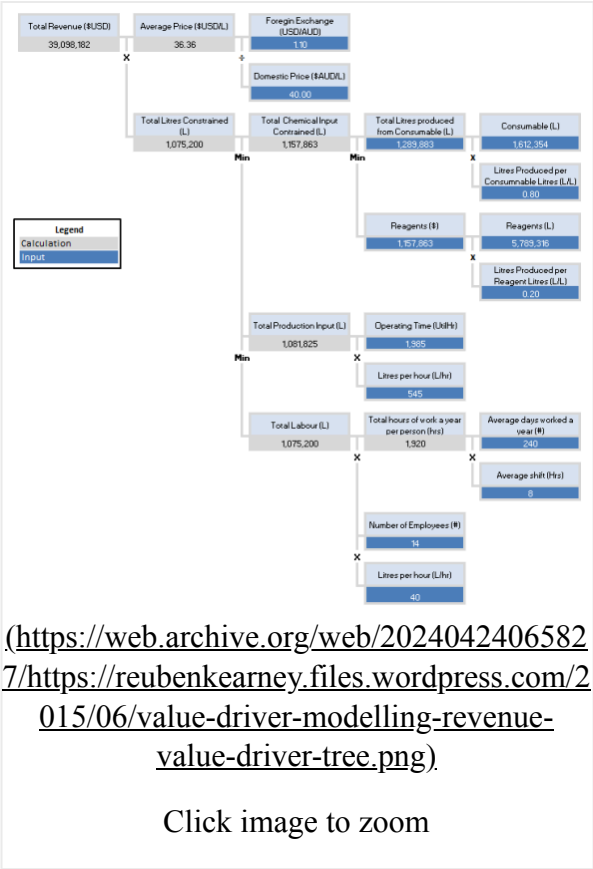


(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-benefits-dependency-value-driver-trees-diagram.png>).

A Benefits Realisation VDT can quantify the benefit as well as show the relationship between the business benefits, operational assumptions and the investment objectives. The matching colours between the above diagrams show where these common elements are.

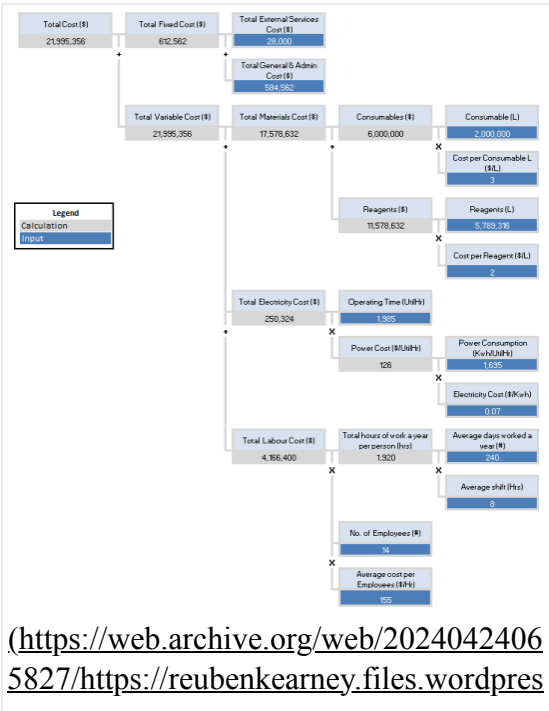
Revenue VDT

Revenue VDTs show the flow of value through the primary activities of an organisation. This model constrains the flow of value based on key inputs. These constraints can show us where improvements could contribute additional value to the organisation. This is an overly simplistic VDT but shows you the basics of what a revenue VDT can do.



Cost VDT

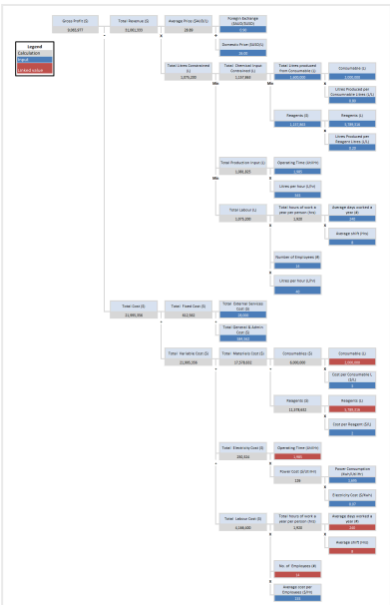
Similar to a Revenue VDT, a cost VDT can use the same inputs to determine what the cost will be to an organisation. This VDT splits costs between fixed and variable. The variable costs are driven by the same input assumptions as the revenue model.



Click image to zoom

A profit VDT is a simple combination of the Revenue and Cost VDTs. This allows us to change a single input and see how this reflects on the overall value to the organisation. This shows the play off between improving productivity and the impact this also has on cost.

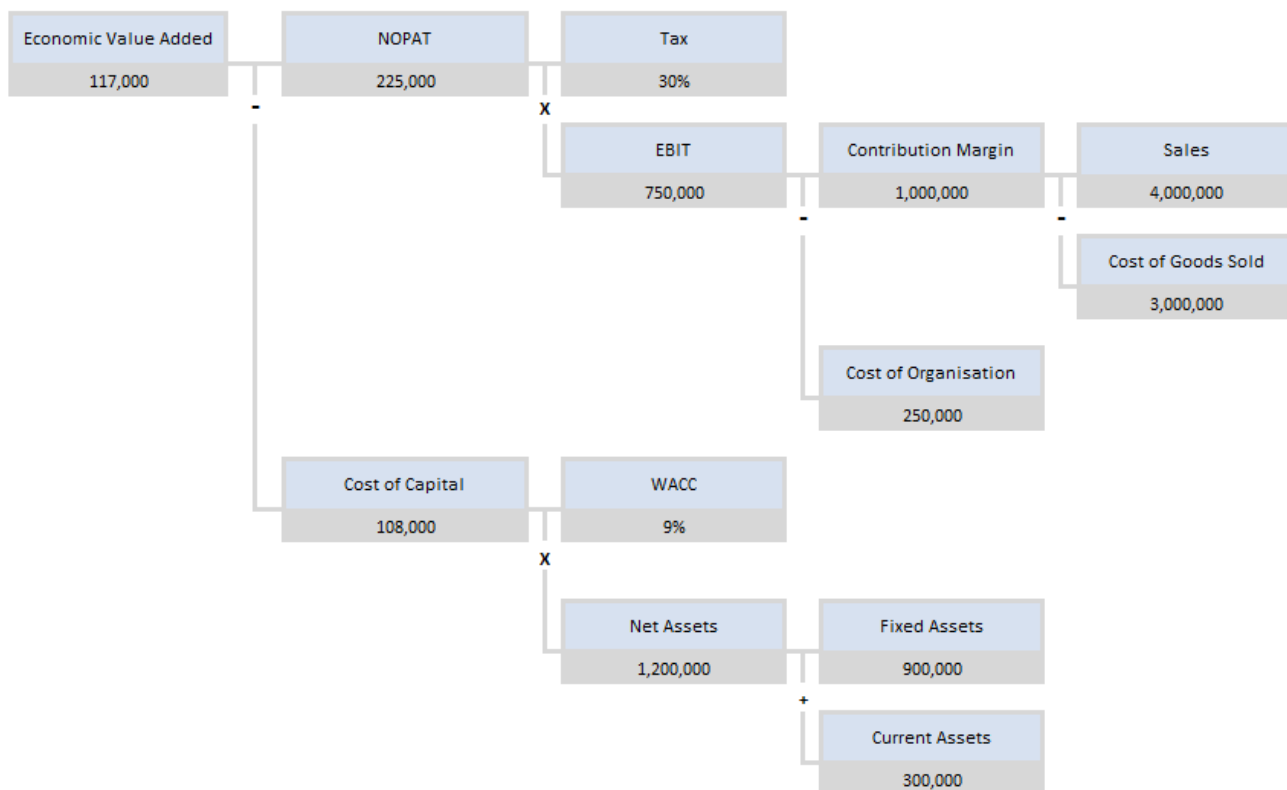
As an experiment, download the [workbook](https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-example-value-driver-trees.xlsx) (<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-example-value-driver-trees.xlsx>) and increase the number of employees working in the organisation from 14 to 15 (at cell O37). You might have assumed that having an additional employee might allow the organisation to create more value (in this case, gross profit). However, as you can see, the additional employee also increases the total cost of labour and without a corresponding increase in productivity for the manufacturing plant itself (see Total Production Input (L) at cell L24), the potential value from the new employee is wasted and you've reduced the organisation's profitability by more than \$100k.



Click image to zoom

Financial VDT

A Financial VDT can be used to assess the financial performance of an organisation. For this example, we are measuring the Economic Value Added or EVA of an organisation. EVA is measure of whether a company is earning better than its cost of capital. These financial inputs can be changed to see how the impact on the EVA of the organisation. Here I have also plugged in some standard accounting ratios to show how you can track the financial performance at the same time.



Name	Definition	Ratio
Asset turnover	sales / assets	3.33
Current asset turnover	sales / current assets	13.33
Fixed asset turnover	sales / fixed assets	4.44
EBIT margin	EBIT / sales	19%
Gross profit margin	(sales – cost of goods sold) / sales	25%
ROA	net income / assets	19%

(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-financial-value-driver-tree.png>).

Reporting VDT

The elements of a VDT can contain whatever information you wish them to. For a reporting VDT we can allocate people to be responsible for specific value drivers as well as show how their achievement is dependent on one another. Here we have cascaded KPIs through the organisational hierarchy to specific individuals. We can report on the status on how they are impacting the creation of value.

For example, in the diagram below, you can see that Susan Grace is doing well by keeping the average cost per employee down, but Luke King is affecting value overall because the average shift is greater than 8 hours. Without this level of detail, you might have only known that Margaret Gold's KPI was on track and not seen the underlying issue of an overworked workforce.

Total Labour Costs (\$)		Margaret Gold (CFO)	Total hours of work a year per person (hrs)		David O'Sullivan (HR)	Average days worked a year (#)		Matthew Wallace (Superintendent)
Planned		2,976,000	Planned		1,920	Planned		240
Actual		2,808,000	Actual		2,160	Actual		240
Variance		-6%	Variance		13%	Variance		0%
						Average shift (Hrs)		Luke King (Supervisor)
						Planned		8
						Actual		9
						Variance		13%
			No. of Employees (#)		David O'Sullivan (HR)			
			Planned		10			
			Actual		10			
			Variance		0%			
			Average cost per Employees (\$/Hr)		Susan Grace (IR)			
			Planned		155			
			Actual		130			
			Variance		-16%			

(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-reporting-value-driver-tree.png>)

Table VDT

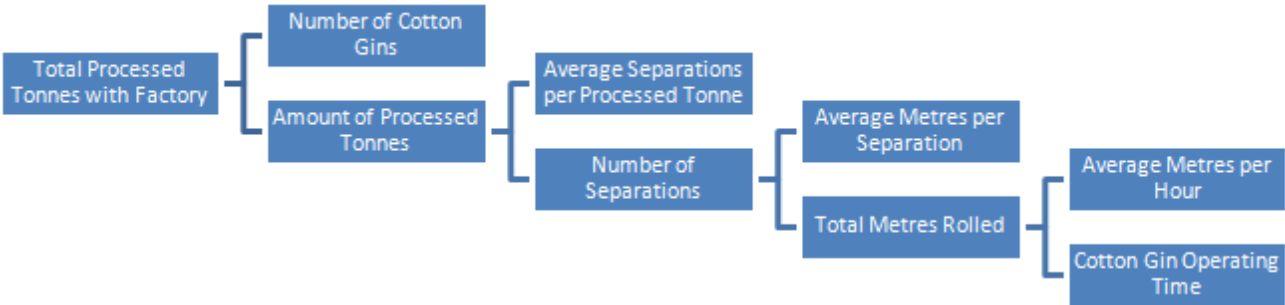
A table VDT contains the same mathematical logic as a VDT however, since it is in a table form, we are able to easily show information over time. In this instance we are using it to measure changes in planned performance. By incorporating time you can analyse seasonable trends or forecast future production.

Parameter	Units	Year 0	Year 1	Year 2	Year 3	Year 4
Cotton Gin Calendar Time (Hrs)		8,760	8,760	8,760	8,760	8,760
Cotton Gin Down Time Maintenance (Hrs)		1,474	1,474	1,474	1,474	1,474
Cotton Gin Scheduled Loss (Hrs)		476	476	476	476	476
Cotton Gin Unscheduled Loss Other (Hrs)		868	868	868	868	868
Cotton Gin Unscheduled Loss Failures (Hrs)		130	130	130	130	130
Cotton Gin Available Time (Hrs)		7,286	7,286	7,286	7,286	7,286
Cotton Gin No Scheduled Production (Hrs)		151	151	151	151	151
Cotton Gin Operating Standby (Hrs)		2,389	2,389	2,389	2,389	2,389
Cotton Gin Utilised Time (Hrs)		4,746	4,746	4,746	4,746	4,746
Cotton Gin Operating Delay (Hrs)		1,093	1,093	1,093	1,093	1,093
Cotton Gin Operating Time (Hrs)	hrs	3,653	3,653	3,653	3,653	3,653
Average Metres per Hour	m/hr	54	54	54	54	54
Total Metres Rolled	m	197,253	197,253	197,253	197,253	197,253
Average Metres per Separations	m/sep	25	24	23	22	21
Number of Separations	sep	7,890	8,219	8,576	8,966	9,393
Average Separations per Processed Tonne	sep/t	300	300	300	300	300
Amount of Processed Tonnes	t	26	27	29	30	31
Number of Cotton Gins	#	10	10	10	12	12
Total Processed Tonnes with Factory	t	263	274	286	359	376
Baseline	t	263	274	314	359	376
Improvement	t	0	0	-29	0	0

(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-table-value-driver-tree.png>).

Click to zoom

If this table was to be represented as a diagram, it would look like the VDT below.



(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-table-value-driver-tree-vdts.png>).

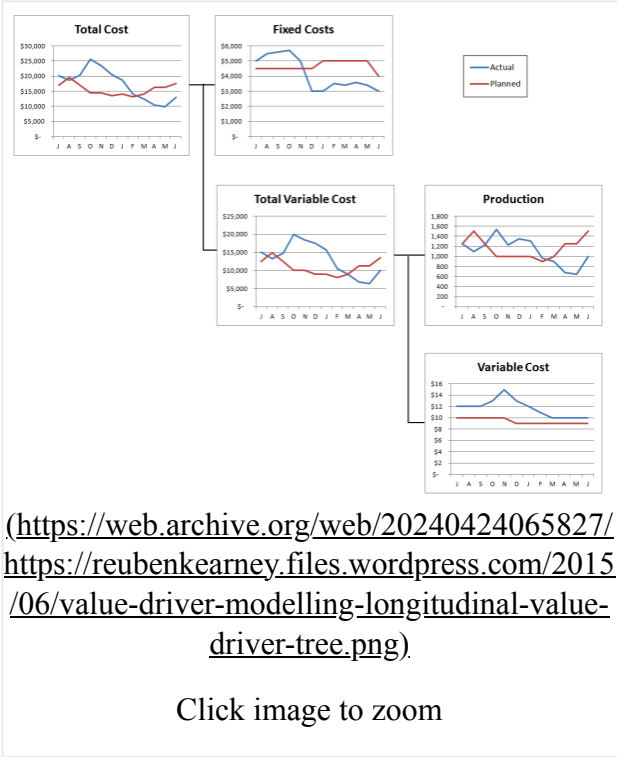
Longitudinal VDT

This final example of a VDT shows the flexibility that VDTs have in presenting and calculating value. In this instance, we are showing the change in value drivers over time. For example, we see that variable costs increased around the same time as production costs. We could hypothesis that, when production increases, economies of scale should see a decrease in costs. So we could focus on this area for investigation

		J	A	S	O	N	D	J	F	M	A	M	J
Production	Actual	1,256	1,100	1,230	1,536	1,234	1,353	1,303	960	900	680	642	1,000
	Planned	1,250	1,500	1,250	1,000	1,000	1,000	1,000	900	1,000	1,250	1,250	1,500
Variable Cost	Actual	\$ 12	\$ 12	\$ 12	\$ 13	\$ 15	\$ 13	\$ 12	\$ 11	\$ 10	\$ 10	\$ 10	\$ 10
	Planned	\$ 10	\$ 10	\$ 10	\$ 10	\$ 10	\$ 9	\$ 9	\$ 9	\$ 9	\$ 9	\$ 9	\$ 9
Total Variable Cost	Actual	\$ 15,072	\$ 13,200	\$ 14,760	\$ 19,968	\$ 18,510	\$ 17,589	\$ 15,636	\$ 10,560	\$ 9,000	\$ 6,800	\$ 6,420	\$ 10,000
	Planned	\$ 12,500	\$ 15,000	\$ 12,500	\$ 10,000	\$ 10,000	\$ 9,000	\$ 9,000	\$ 8,100	\$ 9,000	\$ 11,250	\$ 11,250	\$ 13,500
Fixed Cost	Actual	\$ 5,000	\$ 5,500	\$ 5,600	\$ 5,700	\$ 5,000	\$ 3,000	\$ 3,000	\$ 3,500	\$ 3,400	\$ 3,600	\$ 3,400	\$ 3,000
	Planned	\$ 4,500	\$ 4,500	\$ 4,500	\$ 4,500	\$ 4,500	\$ 4,500	\$ 5,000	\$ 5,000	\$ 5,000	\$ 5,000	\$ 5,000	\$ 4,000
Total Cost	Actual	\$ 20,072	\$ 18,700	\$ 20,360	\$ 25,668	\$ 23,510	\$ 20,589	\$ 18,636	\$ 14,060	\$ 12,400	\$ 10,400	\$ 9,820	\$ 13,000
	Planned	\$ 17,000	\$ 19,500	\$ 17,000	\$ 14,500	\$ 14,500	\$ 13,500	\$ 14,000	\$ 13,100	\$ 14,000	\$ 16,250	\$ 16,250	\$ 17,500

(<https://web.archive.org/web/20240424065827/https://reubenkearney.files.wordpress.com/2015/06/value-driver-modelling-longitudinal-value-driver-tree-table.png>).

The table VDT above, can then be visualised as a diagram as below.



In the next post, I’ll go through some of the fundamental principles you should follow for great VDM design.

[Value Driver Modelling](#) [Value Chain](#), [Value Driver Modelling](#), [Value Driver Trees](#) [9 Comments](#)

9 thoughts on “Value Driver Modelling – Part 1: What are value driver trees?”

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Clear comprehensive examples

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In your Financial VDT, shouldn't the NOPAT be equal to $EBIT \times (1 - \text{Tax})$ rather than $EBIT \times \text{Tax}$?

↳ [Reply](#)

- **Reuben Kearney** says:

[July 12, 2021 at 11:37 am](#)

Correct.

↳ [Reply](#)

- **Josue** says:

[September 17, 2020 at 7:13 am](#)

Longitudinal VDT

“We could hypothesis that, when production increases, economies of scale should see a decrease in costs.”

Why a Decrease in costs?

If I am reading the table correctly, the Total Costs increase when the Production increases. So wouldn't the Hypothesis state there should be an Increase in costs?

↳ [Reply](#)

- **Reuben Kearney** says:

[July 12, 2021 at 11:37 am](#)

Correct. I should have said the ‘cost per unit’ decreases.

↳ [Reply](#)